# AeroSense Developer Manual

Revision: 2026-10-13 R4. Use this manual to assemble, program, and start the reader. Physical and biological validation results are not yet available.

The procedures use ASD-STE100 Issue 9 writing principles. They use short sentences, stable terms, and one action per step. Full vocabulary and rule compliance has not been audited. This document does not claim formal ASD-STE100 compliance. Writing reference.

Use the User Manual for routine controls. Use the USD BOM for purchasing. Use the existing eight parts in 01_print.

## 1 Prepare the parts

Preparation: Use a dry tray, tweezers, scissors, and a 2.5 mm hex key. Complete resin washing and post-curing before assembly. Follow the resin supplier's instructions.

| Part | Quantity | Specification |
|---|---|---|
| 01 base; 02 light shield; 03 gas lid; 08 compression plate | One each | PC/GF-like Black |
| 04 NE; 05 NW; 06 SW; 07 SE wells | One each | Aqua Clear Plus |
| PCB screws | 4 | M3 × 8 mm socket-head cap screws |
| Fan screws | 2 | M3 × 16 mm socket-head cap screws |
| Lid screws | 2 | M3 × 35 mm socket-head cap screws |
| Nuts | 8 | M3 × 0.5 |
| Flat washers | 6 | 3.2 mm inside diameter; 7 mm outside diameter; 0.5 mm thick |
| Fan | 1 | 40 × 40 × 10 mm; 32 mm mounting pitch |
| Seal sheet | As required | 0.8 mm opaque silicone |
| Receiver filter | 1 layer initially | LEE 101 Yellow |

Screw lengths exclude the head. Use socket-head screws, not countersunk screws. The gas shelf is part of lid 03.

Steps:

1. Set SW1 to OFF.
2. Disconnect external power.
3. Remove the battery.
4. Arrange the eight printed parts by their numbers.
5. Separate the three screw lengths.
6. Remove loose support remains and debris.

Keep the parts dry. Do not use a cracked well or damaged wire.



**Historical manual with annotations**

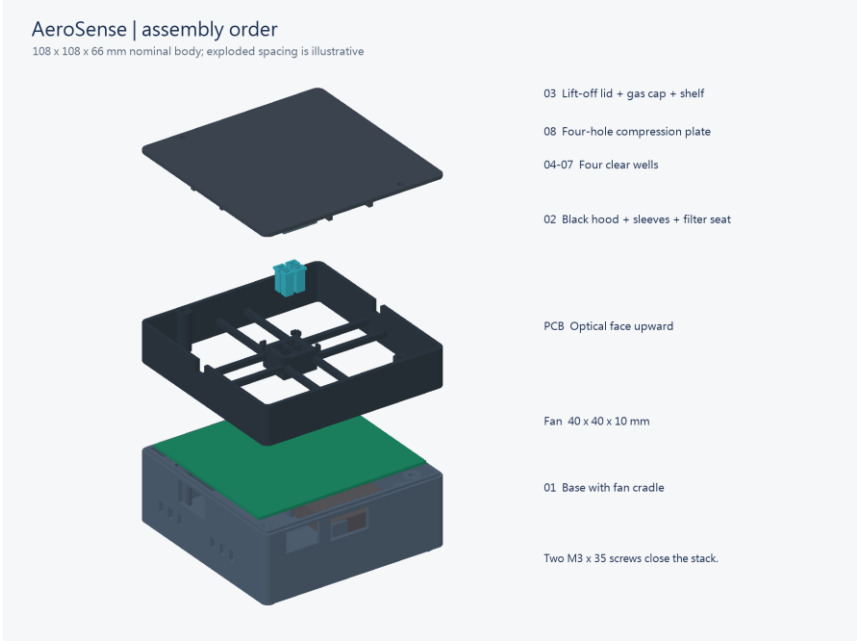
R4 text retained. Review date: 2026-10-19.
Orange: original reviewer notes, anonymized and translated from Chinese.
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**Editorial review · Current fastener correction**

Use M2.5 screws and matching nuts for the PCB. Use M3 × 36 mm screws to close the enclosure. M3 × 8 and M3 × 35 on the historical page are not current purchasing specifications. Verify the PCB screw length against the physical stack.

**Editorial review · Edition scope**

The left page retains the 2026-10-13 R4 record. Separate assembly feedback from optical and biological validation in the revised manual.

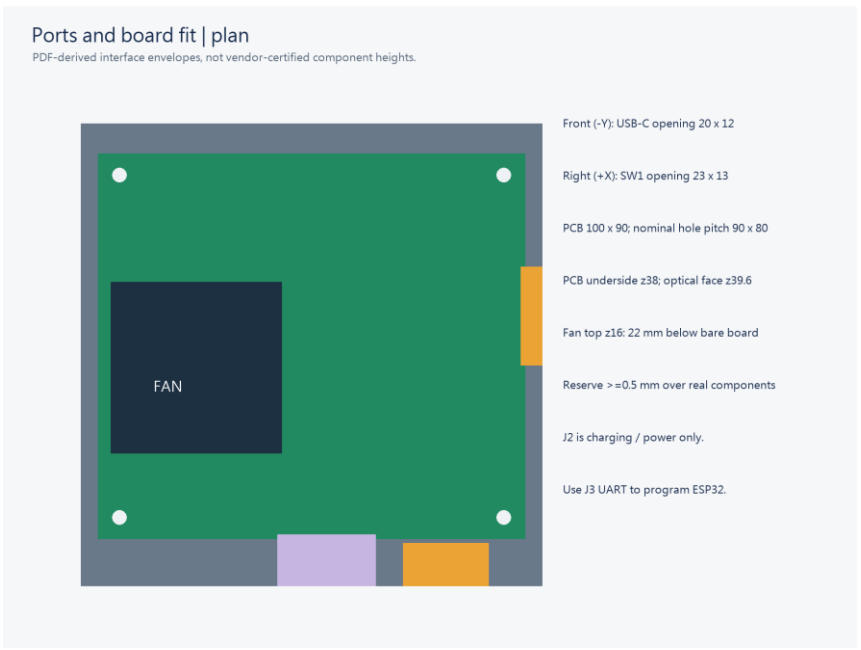



Assembly positions

## 2 Identify the orientation

Put the PCB optical face upward. The USB-C opening is the front. SW1 is on the right when viewed in this orientation. Use the physical parts and the drawing below together.

The well positions are NE, NW, SW, and SE. These names identify fixed assembly positions. Do not exchange them when rotating the reader.




Interfaces and orientation

## 3 Install the fan and PCB

Preparation: Use two M3 × 16 mm screws, four M3 × 8 mm screws, six washers, and eight nuts.

Steps:

1. Put two nuts in the diagonal pockets under the base.
2. Put the fan in its support.
3. Point the fan airflow arrow toward the PCB.



### Historical manual with annotations

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**Original reviewer · §3 Preparation; Chinese p2**

Four 16; two 8.

**Editorial review · Quantities and fan height**

List screws, nuts and washers separately for the fan, PCB and enclosure, with location diagrams. The original 16/8 note is not an approved specification. Raise the fan with a spacer and verify blade, wire and PCB clearance. No exact spacer height is confirmed.

4. Put one washer on each fan screw.
5. Install the two fan screws.
6. Tighten them alternately until the fan is secure.
7. Route the fan wires along the enclosure edge.
8. Put four nuts in the PCB-post pockets.
9. Put the remaining two nuts in the lid-post pockets.
10. Align the PCB antenna with the front opening.
11. Lower the PCB onto its four posts.
12. Put one washer on each PCB screw.
13. Install the four PCB screws loosely.
14. Tighten them gradually in a diagonal sequence.
15. Do not plug the fan into the PCB yet. Use the power connection below.

Check: The fan blades turn freely. The PCB lies flat. Wires and washers do not bridge conductors. USB-C, SW1, J3, SW2, and SW3 remain accessible.

If a part catches, lift it and align it again. Do not force the PCB down with screws. Follow the fan connector's polarity. Never power the fan from an ESP32 GPIO. The fan cools the PCB; it does not supply sample gas.




Fan support and openings

## Fan power

The present fan is labelled DC 5 V, 0.2 A. It has red and black wires with a plug. Nominal input power is about 1 W.

Mounting: if the fan measures 40 × 40 × 10 mm with 32 mm hole spacing, use the base cradle and two diagonal M3 × 16 mm screws. Point its airflow arrow toward the PCB. Do not mount it through unprovided PCB holes.

Power: the schematic has no dedicated fan connector. Use a matching, correctly polarized 5 V USB adapter lead with an external 5 V supply. Allow more than 0.2 A supply capacity and margin for starting current.

Do not connect the fan to J1, J3 or GPIO. J1 connects the battery. J3 carries 3.3 V logic signals. Matching two-pin housings do not establish correct polarity.

Red usually indicates positive and black negative. Confirm this with the fan label and wiring data. Connect fan positive to +5 V and negative to GND. A correct adapter lead avoids soldering.



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**Original reviewer · §3 Step 8; Chinese p3**

They fall out after inserting the PCB. I do not know why they go there. They should go under the base, as with the fan.

**Original reviewer · §3 Step 9; Chinese p3**

The semicircular recess is hard to fit: one barely fits and the other does not. Decided not to use the nuts, but it is loose.

**Original reviewer · §3 Step 10; Chinese p3**

What is the antenna?

**Original reviewer · §3 Step 13; Chinese p3**

The size does not match.

**Original reviewer · §3 Step 14; Chinese p3**

Should the lid be fitted before tightening?

**Editorial review · Nut retention and sequence**

Show the actual nut locations and retention method. Omitting nuts is not an approved solution. Check M2.5 compatibility and the holes, loosely fasten, trial-fit and inspect, then tighten evenly. Label the antenna; distinguish PCB fastening from enclosure closure.

For PCB power, the developer must identify suitable physical 5 V and GND pads. Check supply and starting-current margin first. Do not guess IC or capacitor pins. 5V_USB and boosted 5V are different nets with different switch behavior.

Secure wires with power disconnected. Keep them away from blades, optical windows, tubing and solder joints. Leave the underside intake open. The fan cools electronics; it does not supply sample gas.

Abnormal U2 heating was reported. A fan must not conceal a short circuit, charge-setting fault or supply fault. Do not repeatedly power the affected PCB to test cooling before the fault is resolved.

## 4 Install the receiver filter

Preparation: Start with one LEE 101 Yellow layer. Keep the PCB unpowered. The receiver seat is nominally 5.2 × 4.2 mm.

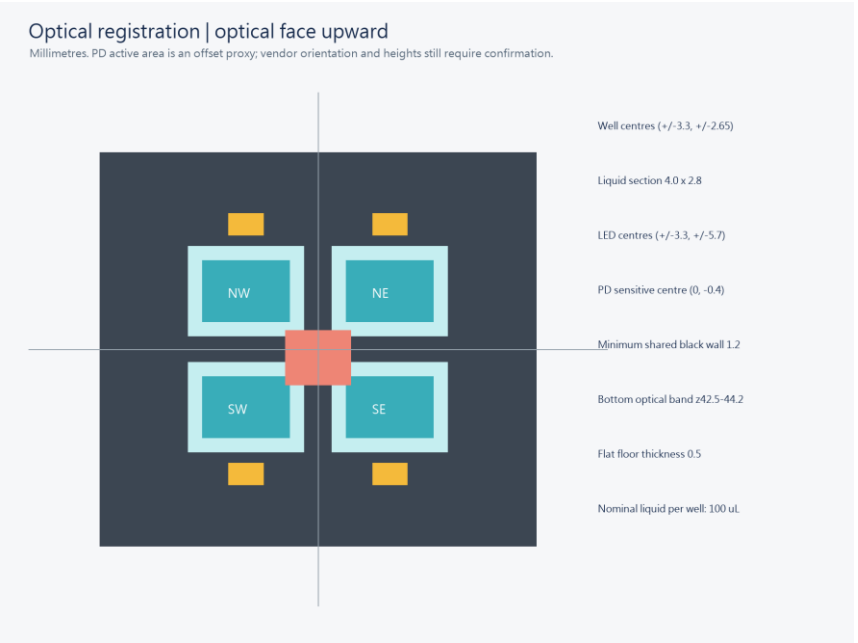
Steps:

1. Cut the LEE sheet to fit the receiver seat.
2. Invert light shield 02 on the dry tray.
3. Put the filter flat in the receiver seat.
4. Cover the PD collection aperture completely.
5. Keep the four LED excitation paths open.
6. Block light paths around the filter edges.
7. Keep adhesive outside the collection aperture.

Check: The filter lies flat. Its edges do not leave an open light path. Do not press or crease the filter.

The experiment compares no filter, one LEE layer, two LEE layers, and one 500 nm OD4 longpass filter. Follow the validation plan for that comparison.

The OD4 candidate is Edmund Optics #29-190. Its 12.45 mm disc does not fit this rectangular seat. Use it only after a suitable mounting method is available. Do not force or arbitrarily cut the disc. Product specification.




Receiver filter and well alignment; 100 µL is the historical model fill.

## 5 Install the seals and wells

Preparation: Open the seal template. Print at actual size, or 100%. Disable fit-to-page scaling. Cut the three seals from the 0.8 mm silicone sheet.



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#### Editorial review · Fan clearance

After adding the spacer, recheck blade rotation, intake openings, wiring and PCB component clearance. Cooling does not establish that the reported U2 fault is resolved; verify the power and overheating issue separately.

#### Editorial review · Middle-layer orientation

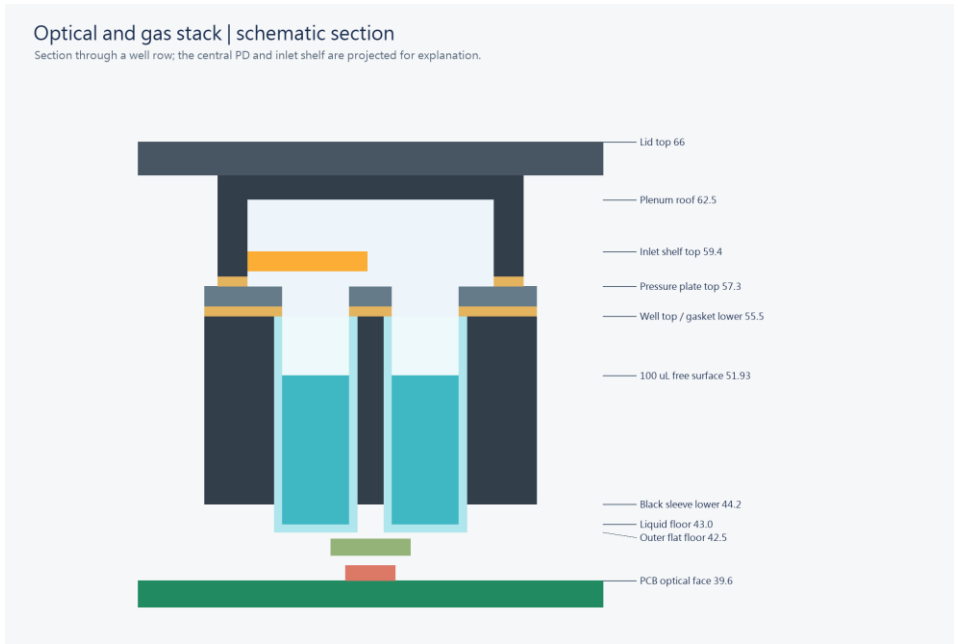
Before lowering the light shield, align the PD collection position with the well centers and check the receiving and excitation paths. Enclosure fit alone does not establish optical alignment. Add matching location marks.

| Seal | Shape |
|---|---|
| PD seal | Small frame around the PD window |
| Lower seal | Four openings aligned with the four wells |
| Upper seal | One rectangular opening above plate 08 |

Steps:

1. Put the small PD seal around the PD window.
2. Keep the active window open.
3. Lower light shield 02 onto the PCB.
4. Insert wells 04–07 in their marked positions.
5. Point each well flange outward.
6. Keep each well floor downward.
7. Put the four-hole seal over the well openings.
8. Align its four holes with the wells.
9. Put plate 08 in its guides.
10. Put the upper ring seal above plate 08.
11. Remove visible debris and wrinkles from the seals.

The wells rest on their flanges. Do not press their floors onto PCB components. Keep plate 08 in the stack. It distributes the lid force across the four wells.




Wells and seal order; 100 µL is the historical model fill.

## 6 Connect the tubes and close the lid

Preparation: Use flexible couplers, gas tubing, two suitable gas filters, and two M3 × 35 mm screws. Leave external gas flow off.

Steps:

1. Find IN and OUT on the gas-lid drawing.
2. Mark these names on the lid.
3. Support the root of the IN tube tail.



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### Editorial review · Enclosure closure

The current enclosure screws are M3 × 36 mm. Seat the washers and seals, keep tubing free, and distribute force evenly. Do not pull misaligned parts together with screws.

4. Push its flexible coupler on gently.
5. Fit the OUT coupler in the same way.
6. Route both tubes through the enclosure openings.
7. Connect the inlet filter and sample tube to IN.
8. Connect OUT through its filter to the downstream suction unit.
9. Keep the outlet open to gas flow.
10. Lower lid 03 over the seal stack.
11. Install the two M3 × 35 mm screws.
12. Tighten them alternately until the lid seats.
13. Keep the tubes free from pinching.

Gas path: sample → inlet filter → IN → chamber → OUT → outlet filter → controlled suction → approved exhaust.

IN is on the CAD −X side. OUT is on +X. The apparent left and right sides change when the lid rotates. Do not force rigid tubing onto printed tails. Do not clamp both ports while closing the lid.

Keep liquids away from the PCB during assembly. The provisional working fill is 80 µL. Approximately 100 µL reaches the reported physical well rim. Before the first wet experiment, follow the fill and leakage check in the sample guide.

The reader does not control gas flow, temperature, CO2, or humidity. Use the airflow model guide for gas-operation context.




Gas shelf and tube connections

## 7 Connect the programming adapter and PCB power

Preparation: Use a USB–TTL adapter (USB–UART) with 3.3 V TX logic. Use a separate USB wall charger labelled 5 V output for J2.

The standard programming route uses two USB connections:

| Connection | Purpose |
|---|---|
| Computer USB → USB–TTL adapter → J3 | Programming signals: TX, RX, and GND |
| Wall outlet → 5 V USB charger → USB cable → J2 USB-C | PCB power |



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The adapter does not power the PCB through these three signal wires. Keep its VCC disconnected. J2 is the power input, not the programming COM interface.

| J3 pin | Signal | Adapter connection |
|---|---|---|
| 1 | 3V3_DIG | Leave disconnected |
| 2 | TXD0, GPIO1 | RX |
| 3 | RXD0, GPIO3 | TX |
| 4 | GND | GND |

Steps:

1. Set SW1 to OFF.
2. Disconnect J2 power.
3. Disconnect the adapter USB cable.
4. Identify J3 pin 1 from the board marking.
5. Set the adapter TX logic to 3.3 V.
6. Connect J3 pin 2 to adapter RX.
7. Connect J3 pin 3 to adapter TX.
8. Connect J3 pin 4 to adapter GND.
9. Leave J3 pin 1 and adapter VCC disconnected.
10. Check for swapped wires, loose metal, and bridged pins.
11. Connect the adapter USB cable to the computer.
12. Connect J2 by USB cable to a charger labelled 5 V output in a wall outlet.
13. Set SW1 to ON.

Check: TX and RX cross. Grounds connect. The PCB has power, and SW1 is ON. A COM port only confirms the adapter connection. With SW1 OFF, the COM port can remain visible while the ESP32 gives no response.

The schematic supports J2 power without a battery. This is a circuit-design conclusion, not a completed physical-board test. With a suitable charged battery at J1 and SW1 ON, J2 is optional.

A VCC jumper does not always change TX voltage. RS-232 and 5 V TX are unsuitable. If the pin order or adapter logic is unknown, keep power off until identified. See the J3 schematic detail.

## 8 Configure and build

Preparation: Install the official PlatformIO IDE extension in VS Code. SW1 is power. SW2 is EN / Reset. SW3 is BOOT.

Steps:

1. Open 03_software/firmware in PlatformIO.
2. If private configuration is absent, copy src/config.example.h to src/aerosense_config.h.
3. Set AP_PASSWORD to at least eight ASCII characters.
4. Set API_KEY to a separate private random value.
5. Keep WIFI_SSID empty for initial direct control.
6. Keep CLOUD_URL empty.
7. Keep BOARD_VALIDATED=false.
8. Save the configuration.
9. Open the PlatformIO terminal.



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10. Run the firmware build command.

```
pio run -e esp32dev
```

11. After SUCCESS, run the filesystem build command.

```
pio run -e esp32dev -t buildfs
```

Check: Both builds show SUCCESS. Firmware and LittleFS are separate outputs. AP_PASSWORD controls Wi-Fi access. API_KEY authorizes webpage commands. Preserve existing private settings.

If WinError 183 names .pio/build/esp32dev, wait for other Build or Upload operations to finish. Then repeat Build once. The cause is not confirmed. A mirror warning can recover if the package later reports installation. Read the final SUCCESS or FAILED result. Do not move the project for this error.

If the build still fails, retain its first error for diagnosis. Do not upload a failed build. Do not publish src/aerosense_config.h or private keys.

## 9 Upload firmware and the webpage

Preparation: Complete Section 7. Keep J2 power connected and SW1 ON throughout both uploads. Find the adapter COM port in Device Manager. COM7 below is an example. Close the serial monitor before upload.

Order: connect wires with power off → supply PCB power → SW1 ON → BOOT/Reset → upload → BOOT/Reset → uploadfs.

Firmware steps:

1. Type this command without pressing Enter.

```
pio run -e esp32dev -t upload --upload-port COM7
```

2. Hold SW3.

3. Press and release SW2.

4. Keep SW3 held.

5. Press Enter in the terminal.

6. After ESP32 detection or Writing starts, release SW3.

7. Wait for SUCCESS.

Filesystem steps:

Keep PCB power on. Repeat the BOOT/Reset sequence for this second upload.

1. Type this command without pressing Enter.

```
pio run -e esp32dev -t uploadfs --upload-port COM7
```

2. Hold SW3.

3. Press and release SW2.

4. Keep SW3 held.

5. Press Enter in the terminal.

6. After ESP32 detection or Writing starts, release SW3.

7. Wait for SUCCESS.

8. Check that SW3 is released.

9. Press and release SW2.

Check: Both uploads succeed. Firmware upload alone does not install the webpage. Use this command when boot output is required:

```
pio device monitor --port COM7 --baud 115200
```

The fixed boot reminder says “upload filesystem before opening UI.” This message can appear after successful upload. It does not prove a filesystem fault. Press Ctrl+C before the next upload.



**Historical manual with annotations**

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If Connecting persists or No serial data received appears, first check J2 power and SW1 ON. Then check COM, crossed TX/RX, common GND, and the BOOT/Reset sequence. A visible COM port does not prove PCB power. If the webpage shows 404, check buildfs and uploadfs.

## 10 Start and check the controls

First resolve U2 overheating. Keep the device dry. BOARD_VALIDATED is a manual setting, not evidence of successful validation.

1. Upload firmware and the web files with uploadfs. Restart the ESP32.

2. Connect the phone to AeroSense Wi-Fi with AP_PASSWORD. Open http://192.168.4.1/.

3. Enter API_KEY. Press Connect. Confirm DEVICE.

4. Select Single measurement. Select All off in Advanced settings. Press Start test.

5. Require 8000 samples, valid=true, railSamples=0 and an empty error field.

6. If only one sample or a sampling error is reported, export JSON and the latest raw CSV. Stop optical interpretation.

7. After this passes, follow User Manual Section 3 10 for covered, weak-light and covered-again measurements.

8. Enable LEDs only after power and ADC checks pass. For the first LED test, select one LED, 5% on-time and one 2 s measurement.

9. Identify the other LEDs in the same way. A fault turns LEDs off immediately. A light that was not visibly observed can still have been commanded briefly.

Interval schedule: 1 minute and 3 measurements targets starts at 0, 60 and 120 s. Each block lasts 2 s. LEDs are off between blocks. The ESP32 schedules them independently of the phone.

Pause discards an incomplete active block and retains the remaining count. Resume waits at least 3 s and retries it. End schedule cancels the remainder. A fault stops the schedule. Restart does not resume it.

Main controls: single or interval measurement, sample name, calibration, start, pause, deletion and history. LED selection, on-time, the fixed 2 s duration and End schedule are in Advanced settings.

Storage: RAM retains the latest 60 summaries until restart. Raw ADC holds only the latest block. Download each CSV before the next acquisition if every waveform is needed.

Compilation and DEMO tests do not prove repaired board timing. The reported one-sample record had a 273 µs read window and adc_read_too_long. This revision retains the existing validity limits.

## 11 Continue with experiments

Assembly and a working interface do not establish optical or cell performance. Use the existing guides below for the scientific procedures.

| Task | Guide |
|---|---|
| Blank, phase reference, dye sequence, filter comparison, and analysis | Validation plan |
| Fluorescein preparation, first wet check, DMEM, seeding, and cell controls | Sample guide |
| Routine controls, acquisition, and data export | User Manual |
| Schedule and accelerated exploratory branch | Completion plan |
| Short independent user trial | Feedback form |



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### Editorial review · Evidence scope

This page retains the R4 interface and firmware procedure. Assembly observations, DEMO tasks and physical-board measurements are separate evidence. This feedback does not establish corrected ADC timing or cellular performance.

Calibration remains necessary. Use the linked experiment procedures before interpreting sample signals. They specify when a changed filter, setting, or assembly needs new calibration.

For handover, provide the Wi-Fi password, API key, firmware version, filter configuration, and actual LED directions. State which experiments have been completed. Cell use requires the sample guide's material and biological checks.

## 12 Open the reader and correct faults

Before opening:

1. Save the session data.
2. Press End schedule and turn LEDs off.
3. Stop external gas flow.
4. Keep the outlet open to equalize pressure.
5. Switch off power.
6. Disconnect external power.
7. Remove the lid screws.
8. Lift the lid, upper seal, plate, and lower seal in that order.
9. Lift each well by its flange.

Keep liquid away from the PCB. Reassemble with Sections 5 and 6.

For interface practice, run this command from 03_software:

```
python server/server.py --demo
```

Open http://127.0.0.1:8765/ on that computer. Confirm DEMO DATA. Use Ctrl+C to stop the server. DEMO values are synthetic.

For an optional private LAN server, use the software setup guide.

| Symptom | Action |
|---|---|
| No programming COM from J2 | Use J3 with the USB–UART adapter |
| Connecting or No serial data received | Check J2 power and SW1 ON first; then COM, TX/RX, GND, and BOOT/Reset |
| Webpage 404 | Build and upload LittleFS |
| 401 | Enter the current API key |
| board_not_validated | Keep Dark until the first electrical checks pass |
| ADC_rail or timing error | Stop and correct the fault before calibration |
| Leakage or abnormal heating | Disconnect power and stop gas flow |
| Network failure prevents stopping | Switch SW1 off |
| upload_queue_full | Restore the server connection before more acquisition |
| no_trained_model_installed | Use acquisition and export; no trained classifier is supplied |

Word is the editable source. PDF is the reading edition. Keep both languages synchronized. Export a new PDF after you edit Word.



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